

C886

Advanced Calculus

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

In a proof-writing seminar that meets online, a student studies $S = \{1 - 1/n : n \text{ a positive integer}\}$. The student's notes are typed in LaTeX. What is $\sup S$, and is it an element of S ?

- A. $\sup S = 1$, and 1 is in S .
- B. S has no supremum because it has no largest element.
- C. $\sup S = 0$, and 0 is in S .
- D. $\sup S = 1$, and 1 is not in S .

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

The sequence $a_n = (-1)^n$, graphed by a teacher in a spreadsheet with 500 terms, is bounded by 1. What can be concluded?

- A. It converges to 0, the average of its terms.
- B. It diverges, and no subsequence converges.
- C. It is bounded but diverges; it has convergent subsequences.
- D. It converges because it is bounded.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which property is guaranteed for every convergent sequence of real numbers?

- A. It is bounded.
- B. Its limit equals one of its terms.
- C. It is monotone.
- D. Its terms are eventually constant.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Proving that the limit of $3x + 1$ as $x \rightarrow 2$ equals 7, a student who has tutored calculus for three years writes "Let $\varepsilon > 0$." Which choice of δ completes the proof most directly?

- A. $\delta = 7\varepsilon$
- B. $\delta = \varepsilon/3$
- C. $\delta = 3\varepsilon$
- D. $\delta = 0.001$, whatever ε is

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

For $f(x) = 5x - 2$, what is the smallest ε for which $|f(x) - 3| < \varepsilon$ is guaranteed whenever $|x - 1| < 0.02$?

- A. 0.06
- B. 0.1
- C. 0.02
- D. 0.004

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A worksheet made with a free graphing tool shows $f(x) = |x - 2|$, which models distance from a mile marker. Which statement about f at $x = 2$ is true?

- A. $f'(2) = 0$ because f has a minimum at 2.
- B. f is differentiable at 2 because it is continuous there.
- C. f is continuous but not differentiable at 2.
- D. f is not continuous at 2 because the graph has a corner.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A function g is known to be differentiable at $x = a$. Which statement must be true?

- A. g has a local extremum at a .
- B. g is continuous at a .
- C. g is differentiable everywhere.
- D. g' is continuous at a .

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A physics student uses $f(x) = 1/x$ on the open interval $(0, 1)$ to model light intensity near a lamp. Which statement is correct?

- A. f is continuous on $(0, 1)$ but not uniformly continuous.
- B. f is uniformly continuous because it is continuous on its domain.
- C. f is not continuous on $(0, 1)$.
- D. f is uniformly continuous because $(0, 1)$ is bounded.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A function f is continuous on $[0, 4]$ with $f(0) = -3$ and $f(4) = 5$; a class checks it on graphing calculators the math department bought last year. What does the Intermediate Value Theorem guarantee?

- A. f is increasing on $[0, 4]$.
- B. There is exactly one c in $(0, 4)$ with $f(c) = 0$.
- C. $f(2) = 1$
- D. There is at least one c in $(0, 4)$ with $f(c) = 0$.

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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10.

A student who plans to teach AP Calculus next fall is asked for the value c guaranteed by the Mean Value Theorem for $f(x) = x^3$ on $[0, 3]$. What is c ?

- A. 1.5
- B. $\pm\sqrt{3}$
- C. $\sqrt{3}$ (about 1.73)
- D. 9

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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11. [TRANSFER — reversed stem / different variable]

A function g is differentiable everywhere, $g(0) = 2$, and $g'(x) \leq 3$ for all x . What is the largest possible value of $g(4)$?

- A. 20
- B. 12
- C. No bound exists without the formula for g .
- D. 14

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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12.

A seminar handout, printed on the department's new laser printer, defines $D(x) = 1$ if x is rational and $D(x) = 0$ if x is irrational, on $[0, 1]$. D is bounded. Which statement is true?

- A. D is not Riemann integrable on $[0, 1]$.
- B. D is Riemann integrable with integral 0 because the rationals are countable.
- C. D is integrable with integral 1 because the rationals are dense.
- D. D is integrable because it is bounded.

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

In a proof-writing seminar ~~that meets online~~, a student studies $S = \{1 - 1/n : n \text{ a positive integer}\}$. ~~The student's notes are typed in LaTeX.~~ What is $\sup S$, and is it an element of S ?

Target word: $\sup S$ and membership **Decides:** Terms approach 1 but $1 - 1/n < 1$ for every n .

A	Confusion set	Treats supremum as maximum.
B	Dropped element	Drops the completeness axiom, which guarantees a supremum.
C	Inverted operation	0 (at $n = 1$) is the infimum, the lower bound.
D	Correct	1 is the least upper bound; no term equals 1.

2. Answer C Concept 2

The sequence $a_n = (-1)^n$, ~~graphed by a teacher in a spreadsheet with 500 terms~~, is bounded by 1. What can be concluded?

Target word: convergence status **Decides:** The terms alternate between -1 and 1 forever.

A	Confusion set	Averaging terms (a Cesàro mean) is not the limit of the sequence.
B	Absolute language	Bolzano-Weierstrass guarantees a convergent subsequence.
C	Correct	Even and odd terms each converge; the whole sequence does not.
D	Inverted operation	Runs “convergent $\Rightarrow$ bounded” backward.

3. Answer A Concept 2 · transfer

Which property is guaranteed for every convergent sequence of real numbers?

Target word: what convergence forces **Decides:** Convergent $\Rightarrow$ bounded.

A	Correct	Past some index terms stay near L ; finitely many others are bounded too.
B	Confusion set	$1/n \rightarrow 0$, and 0 is never a term.
C	Adjacent category	$(-1)^n/n$ converges but is not monotone.
D	Absolute language	$1/n$ converges without ever being constant.

4. Answer B Concept 3

Proving that the limit of $3x + 1$ as $x \rightarrow 2$ equals 7 , a student ~~who has tutored calculus for three years~~ writes “Let $\epsilon > 0$.” Which choice of δ completes the proof most directly?

Target word: δ in terms of ϵ **Decides:** $|3x + 1 - 7| = 3|x - 2|$.

A	Wrong stem number	Builds δ from the limit value 7 , not the slope; it allows $ f(x) - 7 $ up to 21ϵ .
B	Correct	Then $3 x - 2 < 3(\epsilon/3) = \epsilon$.
C	Inverted operation	Multiplies by the slope; gives $ f(x) - 7 < 9\epsilon$.
D	Absolute language	δ must depend on ϵ ; a fixed δ fails for small ϵ .

5. Answer B Concept 3 · transfer

For $f(x) = 5x - 2$, what is the smallest ε for which $|f(x) - 3| < \varepsilon$ is guaranteed whenever $|x - 1| < 0.02$?

Target word: ε from a given δ **Decides:** $|f(x) - 3| = 5|x - 1| < 5(0.02)$.

A	Wrong stem number	Multiplies by the limit value 3 instead of the slope.
B	Correct	$\varepsilon = 5\delta = 0.1$.
C	Dropped element	Drops the slope factor of 5.
D	Inverted operation	Divides δ by 5 instead of multiplying.

6. Answer C Concept 4

A worksheet ~~made with a free graphing tool~~ shows $f(x) = |x - 2|$, ~~which models distance from a mile marker.~~ Which statement about f at $x = 2$ is true?

Target word: continuity vs. differentiability at a corner **Decides:** Left slope -1 , right slope $+1$; no jump.

A	Adjacent category	Fermat's theorem needs f differentiable at the point.
B	Inverted operation	Continuity does not imply differentiability.
C	Correct	The one-sided derivatives differ, but the limit equals $f(2) = 0$.
D	Confusion set	A corner breaks differentiability, not continuity.

7. Answer B Concept 4 · transfer

A function g is known to be differentiable at $x = a$. Which statement must be true?

Target word: what differentiability forces **Decides:** Differentiable $\Rightarrow$ continuous.

A	Confusion set	Differentiability says nothing about extrema.
B	Correct	$g(x) - g(a) = [(g(x) - g(a))/(x - a)](x - a) \rightarrow g'(a) \cdot 0 = 0$.
C	Absolute language	A point property does not spread to every point.
D	Adjacent category	$x^2 \sin(1/x)$ is differentiable at 0 with a discontinuous derivative.

8. Answer A Concept 5

~~A physics student uses~~ $f(x) = 1/x$ on the open interval $(0, 1)$ ~~to model light intensity near a lamp.~~ Which statement is correct?

Target word: uniform continuity on an open interval **Decides:** f blows up as $x \rightarrow 0^+$; no single δ works for a given ε near 0.

A	Correct	Points close together near 0 have values far apart.
B	Dropped element	Drops the closed-and-bounded condition.
C	Confusion set	0 is not in the interval; f is continuous at every point in it.
D	Adjacent category	Bounded is not enough; the interval must also be closed.

9. Answer D Concept 6

A function f is continuous on $[0, 4]$ with $f(0) = -3$ and $f(4) = 5$; ~~a class checks it on graphing calculators the math department bought last year.~~ What does the Intermediate Value Theorem guarantee?

Target word: IVT conclusion **Decides:** Continuous on a closed interval with a sign change.

A	Adjacent category	Monotonicity comes from derivative information, not IVT.
B	Absolute language	IVT gives existence, not uniqueness.
C	Confusion set	Averages endpoint values; IVT says nothing about the midpoint.
D	Correct	0 lies between -3 and 5 .

10. Answer C Concept 7

A student ~~who plans to teach AP Calculus next fall~~ is asked for the value c guaranteed by the Mean Value Theorem for $f(x) = x^3$ on $[0, 3]$. What is c ?

Target word: the point c **Decides:** Average slope $= (27 - 0)/3 = 9$; solve $3c^2 = 9$ in $(0, 3)$.

A	Adjacent category	The midpoint is the MVT point only for quadratics.
B	Dropped element	Drops the requirement that c lie in $(0, 3)$.
C	Correct	$3c^2 = 9$ gives $c = \sqrt{3}$, which lies in $(0, 3)$.
D	Confusion set	That is the slope value $f'(c)$, not c .

11. Answer D Concept 7 • transfer

A function g is differentiable everywhere, $g(0) = 2$, and $g'(x) \leq 3$ for all x . What is the largest possible value of $g(4)$?

Target word: bound on a value from a bound on the derivative **Decides:** MVT: $g(4) - g(0) = 4g'(c) \leq 12$.

A	Inverted operation	Multiplies $(2 + 3)$ by 4 instead of adding 2 to 3×4 .
B	Dropped element	Drops the starting value $g(0) = 2$.
C	Absolute language	The MVT gives a bound from derivative information alone.
D	Correct	$g(4) \leq 2 + 3 \times 4$.

12. Answer A Concept 8

A seminar handout, ~~printed on the department's new laser printer,~~ defines $D(x) = 1$ if x is rational and $D(x) = 0$ if x is irrational, on $[0, 1]$. D is bounded. Which statement is true?

Target word: Riemann integrability **Decides:** Every subinterval holds both rationals and irrationals: upper sums 1 , lower sums 0 .

A	Correct	Upper and lower sums never meet.
B	Adjacent category	That is the Lebesgue integral, a different theory.
C	Confusion set	Density makes every upper sum 1 , but lower sums stay 0 .
D	Absolute language	Bounded is necessary, not sufficient.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Completeness and the supremum	Every nonempty set of reals that is bounded above has a least upper bound; the supremum need not belong to the set.	Treating the supremum as a maximum that must be in the set.	A set like $\{1 - 1/n\}$ that creeps toward a value it never reaches.
2	Convergent implies bounded, not the reverse	Every convergent sequence is bounded; a bounded sequence need not converge, though it has a convergent subsequence (Bolzano–Weierstrass).	Concluding convergence from boundedness alone.	An oscillating sequence such as $(-1)^n$.
3	The ϵ-δ definition of a limit	For every $\epsilon > 0$ there is a $\delta > 0$ (depending on ϵ) so that $0 < x - a < \delta$ forces $ f(x) - L < \epsilon$; for $f(x) = mx + b$, $\delta = \epsilon/ m $ works.	Multiplying by the slope instead of dividing, or fixing δ before ϵ .	“Let $\epsilon > 0$. Choose $\delta = \dots$ ”
4	Differentiable implies continuous	Differentiability at a point forces continuity there; continuity does not force differentiability (corners, cusps).	Running the implication backward.	An absolute value or a corner in the graph.
5	Uniform continuity	A function continuous on a closed, bounded interval is uniformly continuous; on an open or unbounded interval it may not be.	Assuming continuity on any interval gives uniform continuity.	$1/x$ or $\sin(1/x)$ on an open interval touching a blow-up point.
6	Intermediate Value Theorem	If f is continuous on $[a, b]$, f takes every value between $f(a)$ and $f(b)$ at least once.	Claiming exactly one solution, or using it without continuity.	A sign change across a closed interval.
7	Mean Value Theorem	If f is continuous on $[a, b]$ and differentiable on (a, b) , some c in (a, b) has $f'(c) = (f(b) - f(a))/(b - a)$; it also bounds growth from bounds on f' .	Reporting the slope as c , or assuming c is the midpoint.	“Find c guaranteed by...” or a bound on f' .
8	Riemann integrability	Continuous functions (and bounded functions with finitely many discontinuities) on $[a, b]$ are Riemann integrable; boundedness alone is not enough.	Believing every bounded function is integrable.	A function defined differently on rationals and irrationals.

Course scope (public description): Advanced Calculus examines rigorous reconsideration and proofs involving calculus. Topics include real-number systems, sequences, limits, continuity, differentiation, and integration. This course emphasizes students’ ability to apply critical thinking to concepts to analyze the connections between definitions and properties. Calculus III and Linear Algebra are prerequisites.